



LEAD PRODUCT DESIGNER (END-TO-END OWNERSHIP)

CLOUD SOFTWARE GROUP – WEBFOCUS

JAN 2023 – JAN 2025

Train New Model - Binary Classification

Optimize Model



Model Comparison

All Models

Logistic Regression: AUC = 0.9196

K-nearest neighbors: AUC = 0.8223

Random Forest: AUC = 0.9812

Extreme Gradient Boosting: AUC = 0.9924

Neural Networks: AUC = 0.9924

Random Forest: AUC = 0.9079

ROC Precision/Recall

Classification Metrics/Confusion Matrix

Predicted Probabilities

Feature Importances

Learning Curve

Fitted Values

Training Log

Explanation

0.0000 0.1800 1.0000
Favors recall Favors precision

precision = recall threshold

Reset to F1-optimal threshold

Use the slider to modify the threshold. The selected threshold will be saved with the model. The F1-optimal threshold is 0.1800

The Binary Classifiers compute the probability of belonging to the Positive (P) class. A subsequent class assignment requires specifying a decision-threshold. A tradeoff between false positives (FP) and false negatives (FN) is inherent: a small decision-threshold increases the risk of FP and decreases the risk of FN, and vice versa. The effect of the decision-threshold on various metrics is shown in the ROC and Precision-Recall curves. The curves are parametrized by the decision-threshold, which is made visible by the tooltip.

P = Positive | N = Negative TP = True Positive | FP = False Positive | TN = True Negative | FN = False Negative

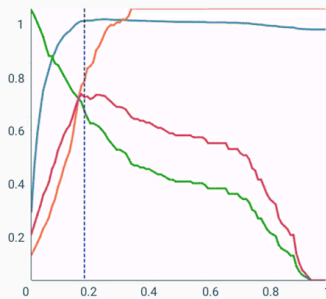
Accuracy: $(TP + TN) / (TP + TN + FP + FN)$ - The correct P and N predictions fraction.

Precision: $TP / (TP + FP)$ - The fraction of P-predictions that are actual P-instances

Recall: $TP / (TP + FN)$ - The fraction of actual P-instances that are predicted to be P-instances

F1-score: $2 * (Precision * Recall) / (Precision + Recall)$ - A Compromise between Precision and Recall

Classification-metrics per threshold



Confusion Matrix

Current Threshold: 0.18

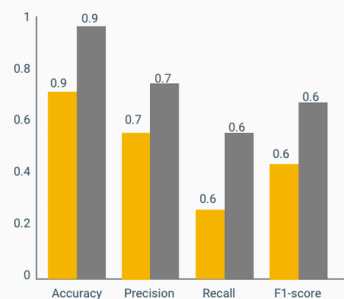
n = 2000	Predicted: N	Predicted: P
Actual: N	TN= 1,811	FP=35
Actual: P	FN=58	TP=96

F1 Optimal Threshold: 0.18

n = 2000	Predicted: N	Predicted: P
Actual: N	TN= 1,811	FP=35
Actual: P	FN=58	TP=96

Target: Attrition Flag
Positive (P) Label: Attrited Customer
Negative (N) Label: Existing Customer
Test set size: 2000 rows

Classification-metrics





EXECUTIVE SUMMARY

NOBODY COULD USE OUR ML ENGINE.

- Anuja Harsha — Senior Product Designer
- ML Functions Case Study

EXECUTIVE SUMMARY

- Situation: A powerful ML engine with zero adoption — buried in a data flow canvas, 12-15 clicks, confusing error states.
- Action: I earned the trust to own this through a small explainability project. Got MIT certified. Designed a 4-step guided workflow with one path serving both novices and data scientists.
- Result: 4/4 SMEs blazed through it without help. Right-click entry pattern adopted for other AI features.

IT WAS REALLY SHITTY.

- Click the plus menu, open a data flow, drag a dataset, drag a model pill.
- Hit a hidden play button. Right-click for cascading context menus.
- "No results generated." 12-15 clicks minimum. Nobody used it.

I DIDN'T DESIGN IT BLINDLY.

- Got MIT certified because the work outgrew surface-level understanding.
- Weekly sessions with the principal data scientist. I validated my understanding constantly.
- If I found this frustrating after weeks of study, a first-time user had no chance.



ONE PATH. NOT TWO EXPERIENCES.

- Step 1: Select Problem Type.
- Step 2: Specify Target Variable.
- Step 3: Choose Predictors.
- Step 4: Configure Hyperparameters.

WHY DON'T WE HAVE A LANDING PAGE HERE?

- No Predict Data landing page existed. I designed it from scratch.
- Tried combining Train and Run. Engineering pushed back. Tab split became the answer.
- Right-click entry: capitalizing on the most common user behavior in the product.

"The popup wizard won because WebFOCUS loved modals. It gave breathing room for helper text. Sometimes the best design move is the one that doesn't ask the legacy product to become something it isn't."

THE MOST COMPLEX SCREEN I'VE EVER DESIGNED.

- Confusion matrix: two charts, a threshold slider, a four-by-three table.
- Threshold slider and chart point: exactly 90 degrees parallel.
- A dozen sessions with the data scientist — paper, pen, sketching live inside my design software.

THEY JUST BLAZED THROUGH IT.

- 4/4 SMEs completed the workflow without help.
- "The best screen in the entire UX revamp." — the principal data scientist said it out loud, in front of everyone.
- Right-click entry pattern adopted for Generate Insights and Ask a Question.

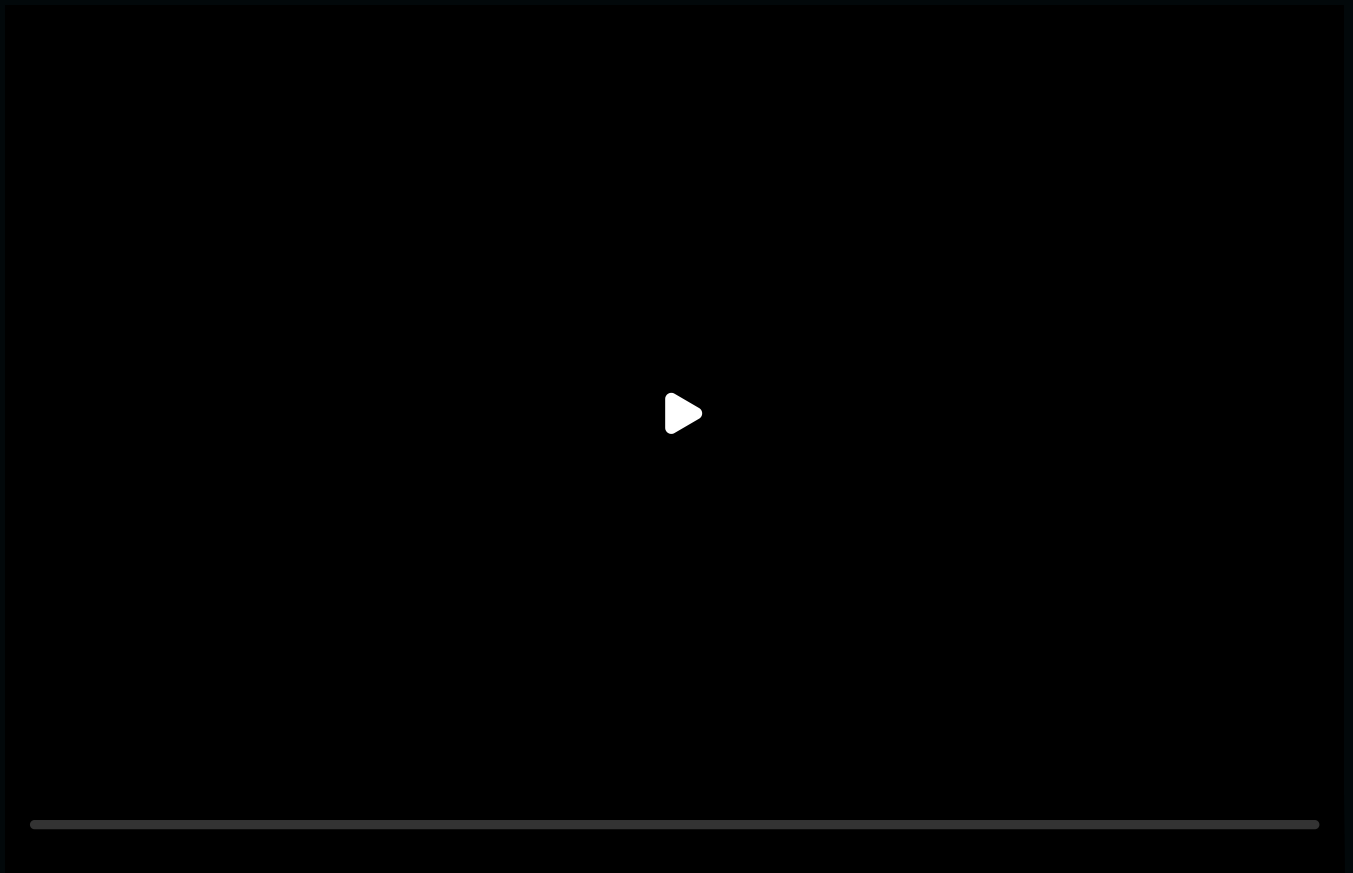


WHAT A MIND-BENDING PROJECT.

- ML taught me product partnership. My PM and I co-owned the direction.
- I fought for every decision with logic, research, and documentation.
- I only fully realized I was shaping the entire product experience after getting laid off.



ML FUNCTIONS REDESIGN • WORKFLOW WALKTHROUGH WITH VOICEOVER





A Side Project Earned Me the Whole Thing.

While deep in ReportCaster, the principal data scientist handed me a side challenge — an explainability popup. I nailed it, and that earned the trust to own the entire ML workflow redesign. Here's what I walked into, and how it all started.

MIT

DOMAIN MASTERY

Zero ML background. I asked my data scientist questions. I asked AI questions. I researched binary classification, regression, random forest, neural networks — whatever I needed. Eventually that pushed me into taking an MIT course. I paid for it myself because the work had outgrown surface-level understanding.



ZERO ADD-ON

~15 clicks minimum just to see a trained model. Plus menu → data flow → drag dataset → drag model pill → hidden play button → cascading context menus for hyperparameters. Powerful tool. Nobody used it.

THE SIDE PROJECT THAT STARTED IT ALL

The screenshot displays the 'WebFOCUS - Explainability Popup' interface. On the left, a sidebar titled 'Turn Prediction Tool: TRAINING PHASE with Feature Importances and XWIN Batch Explanation' shows a list of predictors for a 'bankchurners.csv' dataset. The main panel, titled 'Conventional Feature Importances', displays a horizontal bar chart showing the importance of various features. The top features are 'Total_Trans_Ct', 'Total_Trans_Amt', and 'Total_Relationship_Count'. On the right, a section titled 'with the values and see how the model outcome changes' shows the impact of modifying specific features. It includes a table with 'Total Revolving Balance' and 'Total Count Change', and a 'Recompute model outcome' button. Below this, a comparison of 'Modified Model Outcome' (0.4) and 'Actual Model Outcome' (0.38) is shown. The bottom right corner of the popup shows a list of regions: North America, South America, EMEA, and North America.

WHAT THE DATA SCIENTIST SHOWED ME

WHAT I BUILT

I dug into why the popup was needed, how model predictions work, what feature importances actually mean. He was glad I took the effort in understanding.

▶ [Watch public explainability demo](https://www.youtube.com/watch?v=oPFKkcgNCbo) (https://www.youtube.com/watch?v=oPFKkcgNCbo)



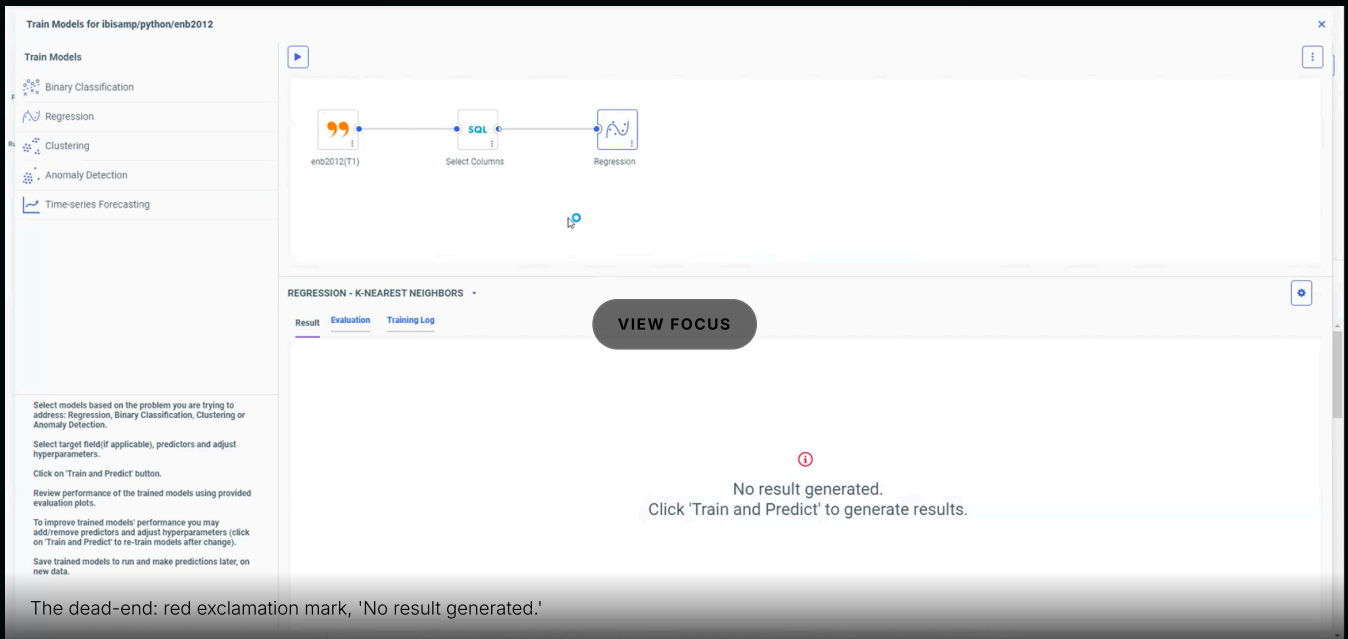
"The other designers I worked with before didn't really understand what they were doing — they just gave designs. But you sat with us, talked to us, and actually understood. That's why I trust you."

— MARCUS HORRACH, PRINCIPAL DATA SCIENTIST

LEGACY PAIN POINTS

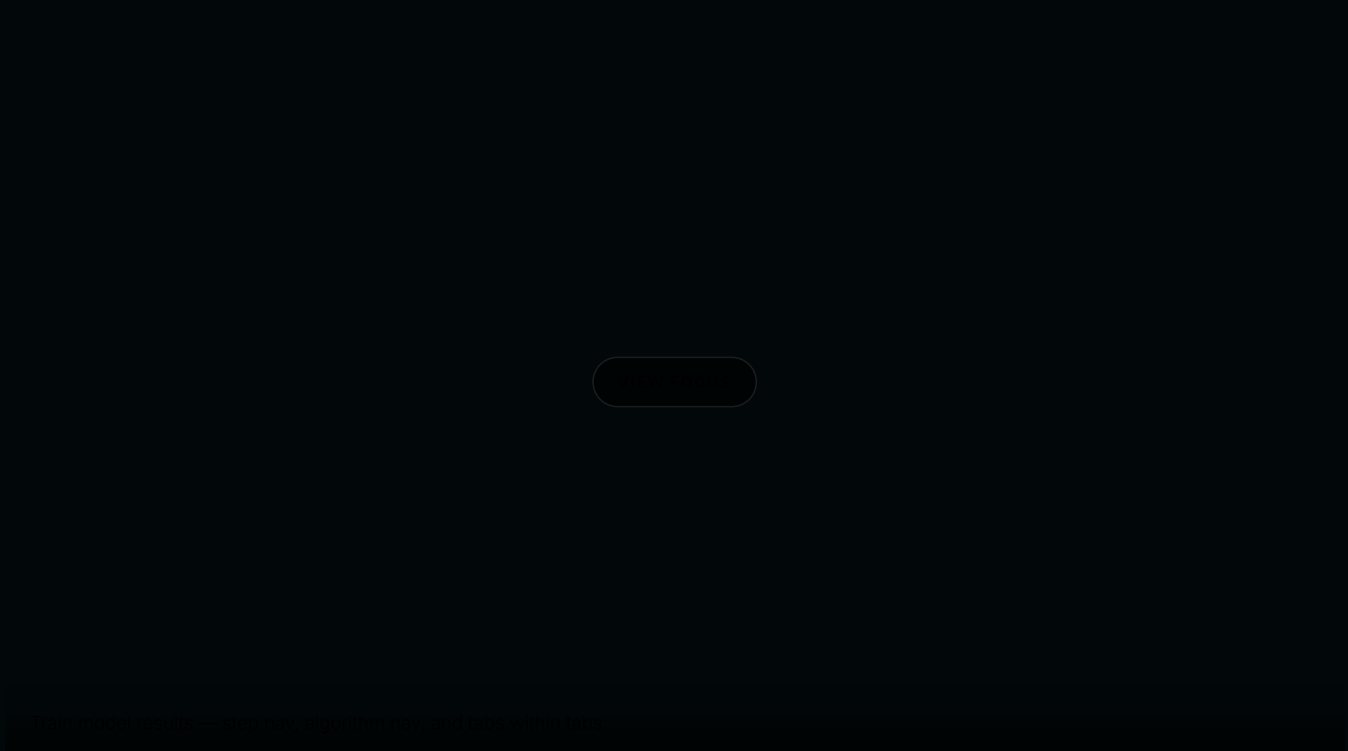
"No Result Generated."

There was a big exclamation mark in a red circle, and the expression was red itself. It said 'No model generated, please click on the play button to start training the model.' So frustrating. That was the first thing I noticed, and it was the first thing I said — let's remove that messaging, please.



Three Levels of Navigation.

You had the step navigation, you had the algorithm navigation, and inside algorithms, another level of tabs. Just for binary classification — 8 into 5 screens. That same structure applied to every model type.





The Data Flow Canvas.

The old ML workflow had the data flow canvas visible at all times. Split view — data flow on top, table view at the bottom. You didn't really need the data flow in the view. I had to ask the data scientist for repeated walkthroughs because I couldn't get the sandbox to work.

view focus

Run Model: data flow canvas on top, side data panel. Everything interconnected.

THE TURNING POINT

It was also something I had proposed — because after I found out about the explainability popup, I realized the entire machine learning workflow was not great. Understanding earned me ownership.



One Path. Not Two Experiences.

Data scientists wanted a technical workflow. I wanted something business users could actually use. The principal data scientist at one point was like, "Are you trying to make this dumb?" No. I wasn't removing sophistication. I was making the sophistication legible. We landed on one workflow that serves a dual purpose — not dumbed down, not over-engineered.

2

SESSIONS WITH THE DATA SCIENTIST

We used to talk about physics and books. I used to show him sketches — "Am I understanding this right?" I validated my understanding constantly because I was terrified of making something look nice while technically meaning the wrong thing. That would have been a fake win.

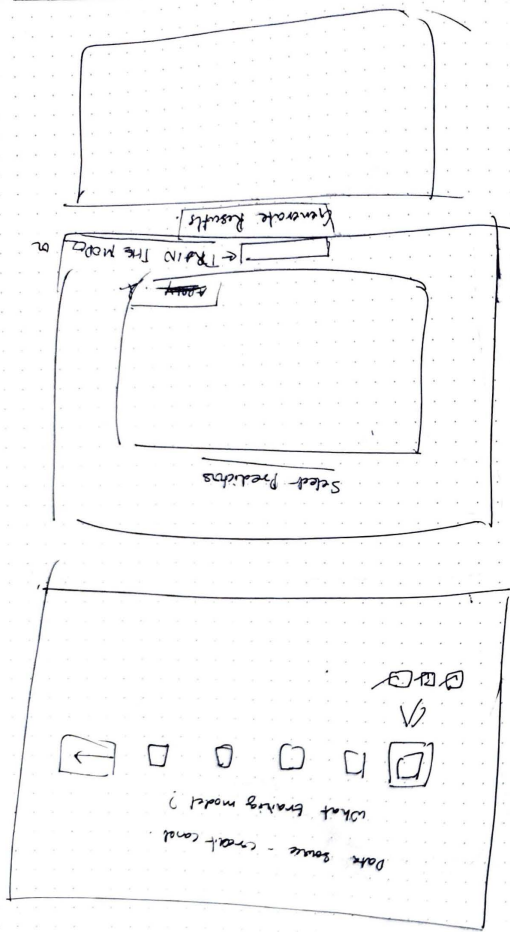


PUSHING BACK

We had this conversation about chart colors — four different colors that didn't mean anything. I said, "Your goal is to show F1 scores. Adding colors makes it more confusing. You just want users to compare and pick the best one." I won that conversation because I'd finally understood what the chart was actually showing.

SKETCHBOOK SESSIONS

- go to application directory
- Right click on a DSC file and select "create flow"
- Click on Train Models tab in the sidebar
- select a model to train
- Dialog box opens up for configuring "Clustering"
- Click on apply — empty page appears
- Click on "Play" button which is Train & Predict.
- Then there is a 2 min training time
- Result appears in a table in a quarter of the screen. as a dialog box with different bar charts
- Then using a save button the user can save a trained model.
- Then refresh and you see a "RunModel" tab in the side bar.
- select the same trained model to run on the data set of your choice.
- An icon appears in your data flow canvas when you can right click and change & edit settings and run it (copy) from the dialog.





[VIEW FOCUS](#)



PERSONA RESEARCH

- Data Scientist → Depth & Control
- Business Analyst → Simplicity
- Tech Analyst → Both

We did not have two experiences. We had one path that serves a dual purpose. Not easy enough that data scientists would say “this is spoon feeding,” but clear enough that someone like me could use it. Every step had context — definitions, explanations, tooltips. The sophistication was there; it just waited for you to ask for it.

ARCHITECTURE & WIREFRAMES

view focus

Initial hand-drawn wireframes



view focus

12 column grid layout

view focus

Full workflow mapping



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All model types architecture

VIEW FOCUS

High-fidelity UI Structure

THE TURNING POINT



Every session with the data scientist was enlightening. I don't think without those sessions I would have been able to do it at all. He loved machine learning. I loved user experience. That mutual passion is why it worked.

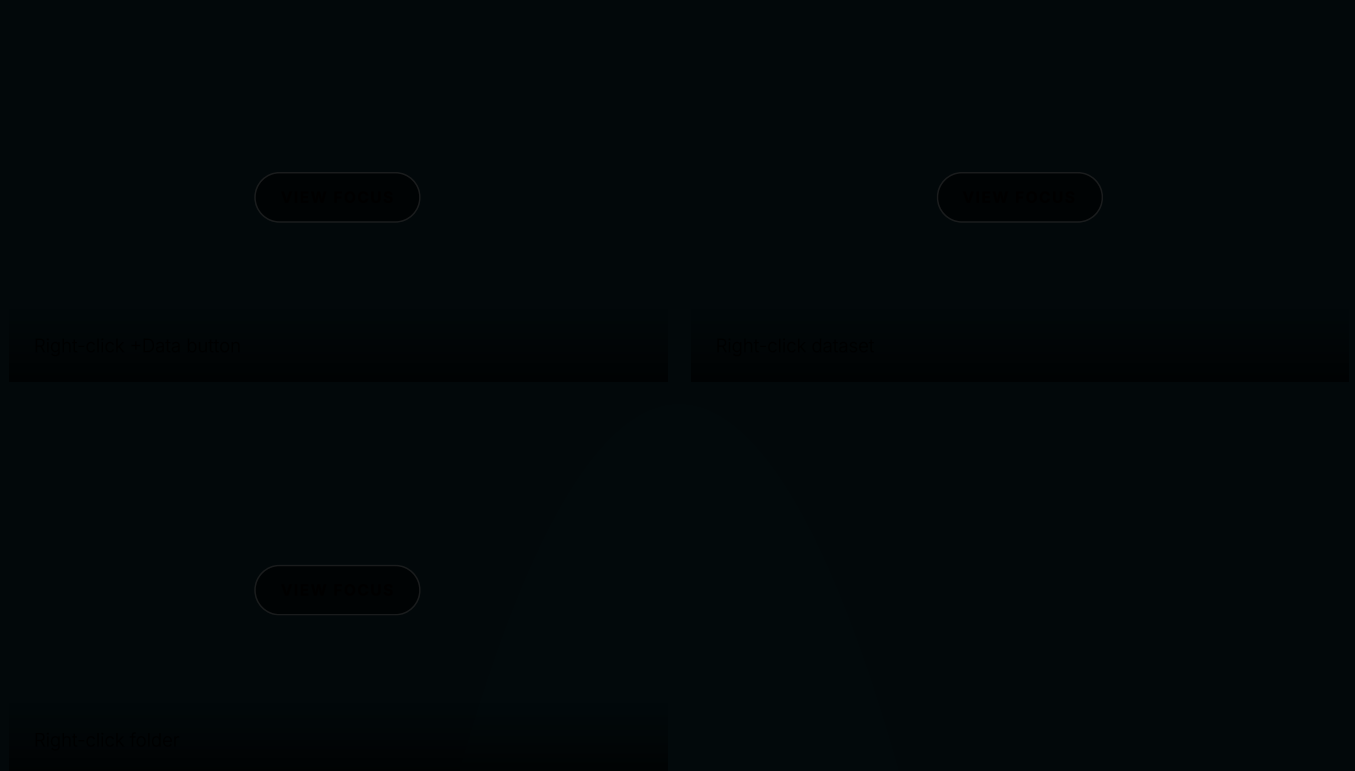




Why Don't We Have a Landing Page Here?

No Predict Data landing page existed. My first instinct was — why can't users select data themselves? I designed it from scratch, dozens of iterations, until the tab split became the answer.

ENTRY POINT DESIGN



Six Clicks to Two.

In WebFOCUS, right-click is like religion. Everything is right-clickable. So putting “Predict Data” in the right-click menu of a dataset was the most natural thing. I was capitalizing on the most common user behavior in the product. Once we saw how well it worked, we extended the pattern to Generate Insights and Ask a Question too.

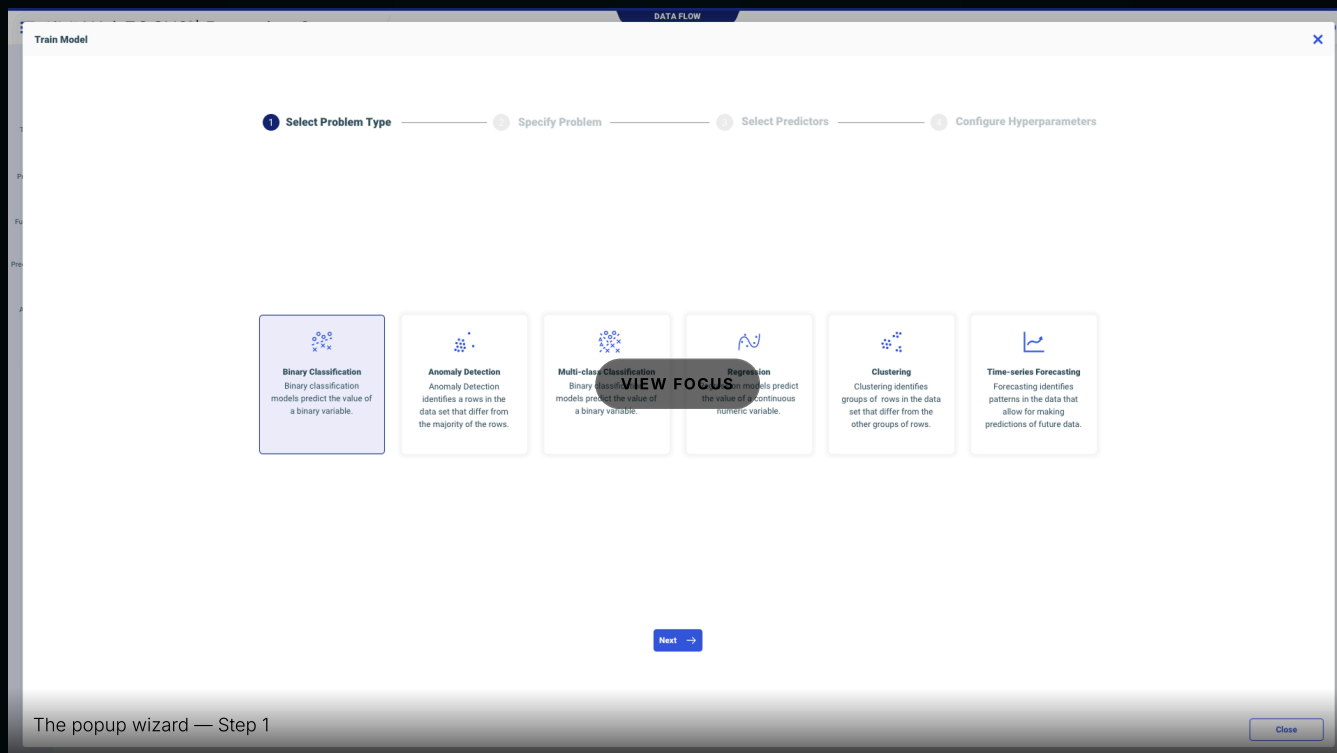


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Initial workflow mapping

The Popup Won.

I tried full-screen views, side-step layouts, two-column layouts. The popup wizard won because WebFOCUS loved modals — all engineers lived in popups. It gave me breathing room for helper text and definitions. It let each step exist cleanly without the whole screen jumping. And it was faster to engineer. Sometimes the best design move is the one that solves the problem without asking the legacy product to become something it isn't.

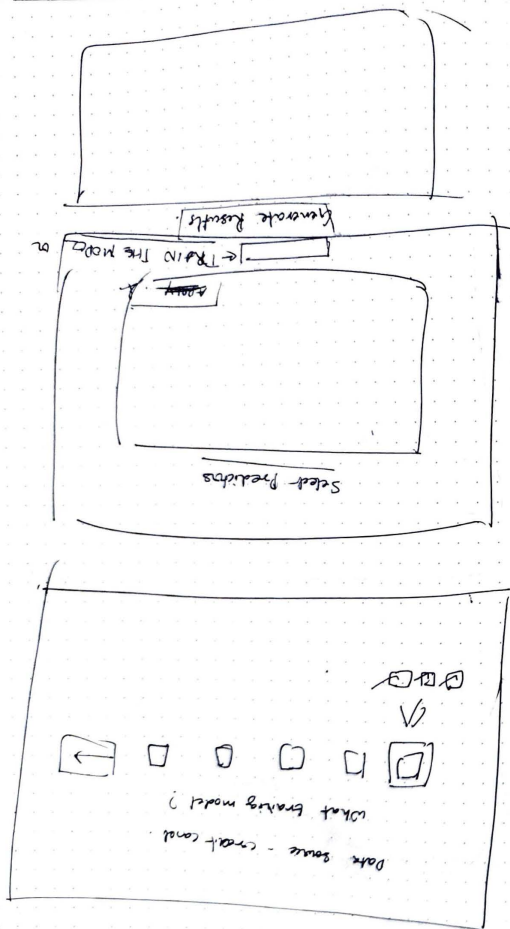


The popup wizard — Step 1

Close

SKETCHES & PROCESS

- go to application directory
- Right click on a DSC file and select "create flow"
- Click on Train Models tab in the sidebar
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- Dialog box opens up for configuring "Clustering"
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[View more](#)







The Most Complex Screen I've Ever Designed.

The confusion matrix screen had two charts, a threshold slider, and a four-by-three table. When I first saw it, I couldn't make sense of it. So I sat with the data scientist — a dozen sessions, paper and pen — until I understood every single interaction.

THE GUIDED 4-STEP WORKFLOW

VIEW FOCUS

Entry point: Empty state with clear CTA



DATA FLOW

Train Model

1 Select Problem Type

Specify Problem

Select Predictors

Configure Hyperparameters

Binary Classification
Binary classification models predict the value of a binary variable.

Anomaly Detection
Anomaly Detection identifies a rows in the data set that differ from the majority of the rows.

Multi-class Classification
Binary classification models predict the value of a binary variable.

Regression
Regression models predict the value of a continuous numeric variable.

Clustering
Clustering identifies groups of rows in the data set that differ from the other groups of rows.

Time-series Forecasting
Forecasting identifies patterns in the data that allow for making predictions of future data.

Next →

Close

Step 1: Select Problem Type

VIEW FOCUS

Step 2: Specify Problem Details



VIEW FOCUS

Step 3: Select Predictors

VIEW FOCUS

Step 4: Configure Hyperparameters

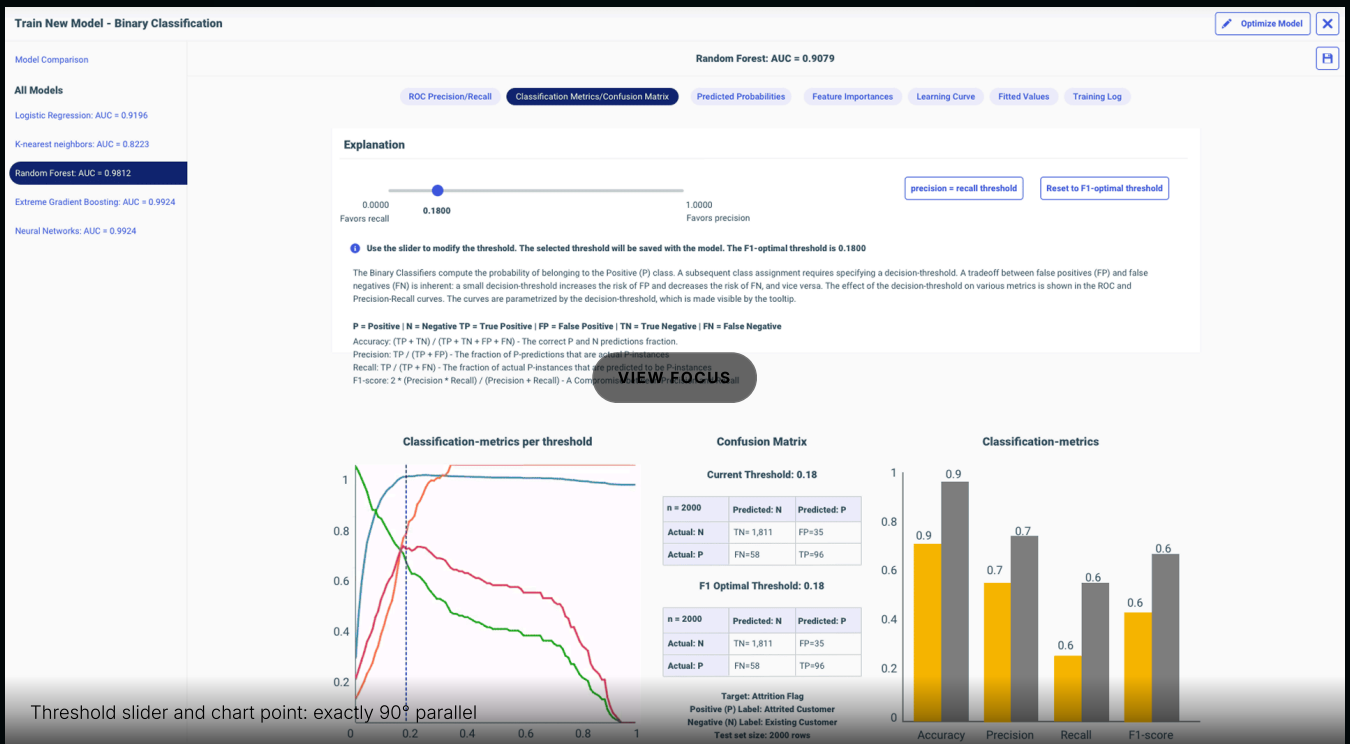


VIEW FOCUS

Compare Models — side-by-side evaluation

The Confusion Matrix

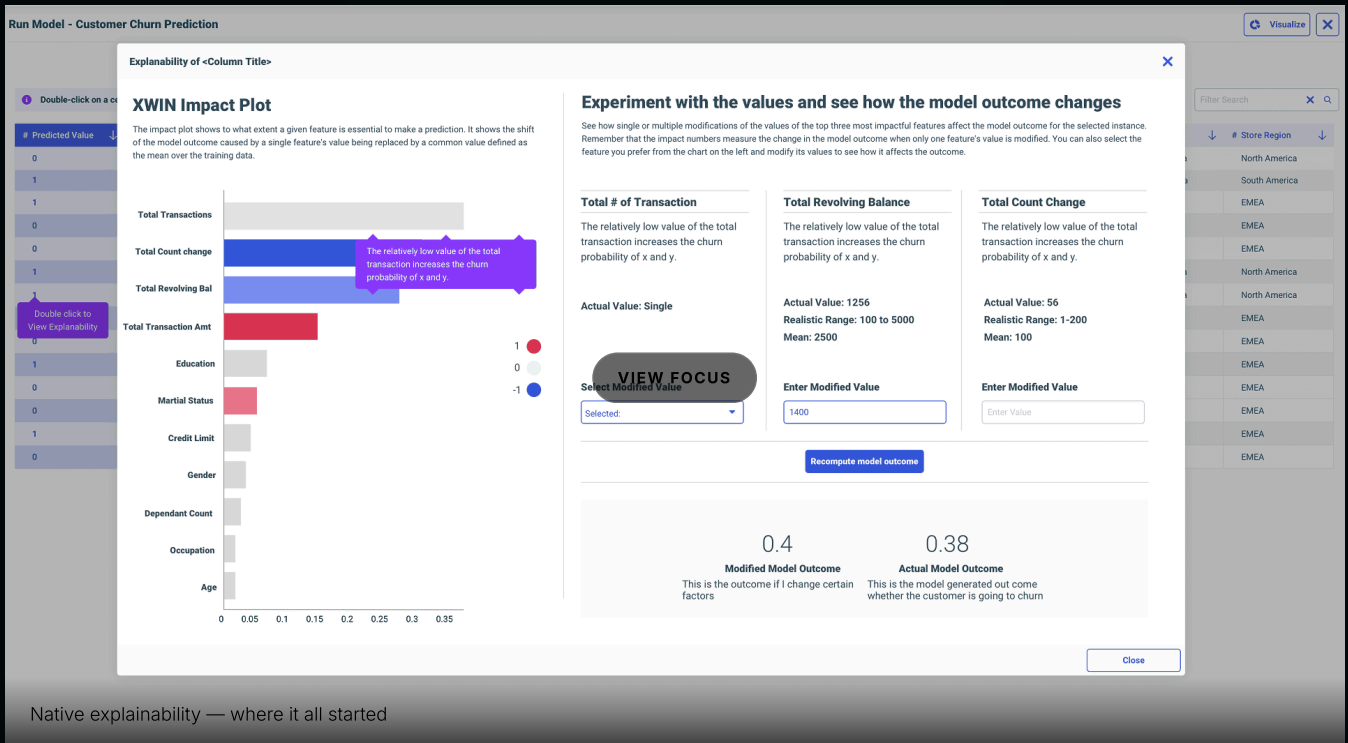
I aligned the threshold slider directly under the chart point that moved — exactly 90 degrees parallel. As you drag it, the dot on the line chart moves with it. There's no way you can miss the relationship. It was like magic. After understanding every interaction on every chart, I could finally explain it to somebody else visually.





VIEW FOCUS

ROC / Precision curves



Native explainability — where it all started



view focus

Optimize Model — with more warnings

view focus

Full ML UI structure

STYLING & STRUCTURE DECISIONS



view focus

12 column grid system for wizard steps

view focus

Grid applied to training UI



VIEW FOCUS

Model tile card spec

VIEW FOCUS

Table styling specification

BEFORE / AFTER WALKTHROUGH

LEGACY WORKFLOW



Train Models for ibisamp/python/enb2012

Train Models

- Binary Classification
- Regression
- Clustering
- Anomaly Detection
- Time-series Forecasting

Select models based on the problem you are trying to address: Regression, Binary Classification, Clustering or Anomaly Detection.

Select target field (if applicable), predictors and adjust hyperparameters.

Click on 'Train and Predict' button.

Review performance of the trained models using provided evaluation plots.

To improve trained models' performance you may add/remove predictors and adjust hyperparameters (click on 'Train and Predict' to re-train models after change).

Save trained models to run and make predictions later, on new data.

enb2012(T1) → Select Columns → Regression

REGRESSION - K-NEAREST NEIGHBORS

Result Evaluation Training Log

No result generated.
Click 'Train and Predict' to generate results.

REDESIGNED WORKFLOW

Application Directories

Get Data Filter Manage

Applications

Search directories

Application Directories/Files	Size	Date Modified
oefiles	5466	2020/12/29 09:27:44
prod	987	2020/12/29 09:27:45
rpcat	40755	2020/12/29 09:27:44
zcomp	4668	2020/12/29 09:27:44
car_df	5074	2023/04/03 06:26:32
brokers	653	2020/12/29 09:27:41
carolap	3240	2020/12/29 09:27:45
cashflow	1463	2020/12/29 09:27:42
country_population_2014	1482	2022/06/24 07:57:51
course	964	2020/12/29 09:27:44
courses	665	2020/12/29 09:27:44
educfile	677	2020/12/29 09:27:43
empdata	1155	2020/12/29 09:27:44
employee	2398	2020/12/29 09:27:43
expension	1458	2020/12/29 09:27:44
finance	456	2020/12/29 09:27:42
ggdemog	1555	2020/12/29 09:27:42
ggorder	992	2020/12/29 09:27:42
ggprods	1031	2020/12/29 09:27:42
ggstores	935	2020/12/29 09:27:42
hday	473	2020/12/29 09:27:44
ibimvdm	1318	2020/12/29 09:27:44
ibimvdm	898	2020/12/29 09:27:44
ibimvgr	877	2020/12/29 09:27:44

Open
Get Insights
Create Visualization
Predict Data
Sample Data
Data Profiling
Analysis Reports
Metadata Management
Data Management
Create Data Flow
Copy
Delete
Cut
Rename
Properties

ibisamp/python/enb2012 Legacy_ui.exe workflow_step_1.tsxx

enb2012(T1) → Select Columns → Regression

REGRESSION - K-NEAREST NEIGHBORS

Result Evaluation Training Log

1 Select Predictors 2 Configure Hyperparameters

Regression
Regression models predict the value of a continuous numeric variable.

Clustering
Clustering identifies groups of rows in the data that are different from the others.

Time-series Forecasting
Forecasting identifies patterns in the data that allow for making predictions.



Based on the problem you are trying to solve, select a machine learning algorithm (e.g., Binary Classification, Clustering or Regression).
Select applicable predictors and adjust their weights.
Click the 'Predict' button.
View the predictions of the trained models using provided data.
Compare the models' performance you may adjust hyperparameters (click 'Re-train' to re-train models after change).
Click 'Run' to run and make predictions later, on new data.

LEGACY

Click



TRAINING: PREPARE DATA

GET MODEL WEIGHTS FROM OTHER GROUPS OF ROWS.

ENSURE OUR TRAINING PREDICTIONS OF FUTURE DATA.

REDESIGN

THE BEST SCREEN IN THE ENTIRE UX REVAMP.

**"The best screen in the entire UX revamp."
He said that during a team walkthrough.
Out loud. In front of everyone. — Marcus
Horbach, Principal Data Scientist**



I Fought for Every Damn Thing in ML.

I was the only designer on ML, on IQ, on Reporting Server. After the layoffs, it became four systems. I fought for every design decision with logic, not instinct.

RC

ML

IQ

WD

I was the only designer to work on Reporting Server, on ML, on IQ. Nobody else touched these. I built a mental model of WebFOCUS as an end-to-end ecosystem: get data, prepare data, train and run ML, visualize in Designer, distribute through ReportCaster. Nobody explained that to me. I built it by working across enough of the system.



THE DATA FLOW FIGHT

When the data scientist tried to bring Data Flow back into the view, I pushed back. I had recordings, notes, documentation from our earlier sessions. I reminded him of our prior decisions, explained why the change would hurt the workflow, and held my ground. The room was me, two directors of engineering, the head PM, the principal data scientist, and my design director. I won.

Defending my thinking forced me to depend on logic, not instinct. I always needed industry research, existing WebFOCUS behavior, documentation, or common sense to back up my decisions. I could never just say "I designed this because I felt like it." Design instincts don't mean anything if you don't have the evidence to back them up. That made me sharper. That made me better.



They Just Blazed Through It.

Four SMEs. Separate sessions. I told them one thing: “It starts from the dataset.” They right-clicked, saw “Predict Data,” clicked it — and just blazed through the whole workflow. Because it was that obvious. The four-step flow was untouched by feedback. They had design suggestions on the results screens, but the core spine? Nobody questioned it.

4/4

Every SME completed the guided flow without assistance. My design director, who knew nothing about ML, was able to do it too. I taught him machine learning with this workflow. That was the real test.

~6 → 2

Clicks to start. Right-click, Predict Data, done. The entry point I designed became how other AI features surfaced in the same menu — Generate Insights, Ask a Question. That wasn't just a UX tweak. That was adoption thinking.



What a Mind-Bending Project.

ML Functions taught me product partnership more than anything else. With RC, I had to dig through chaos. With ML, my PM and I co-owned it. My input shaped the roadmap, the concepts, the direction. I only fully realized I was shaping the entire product experience after getting laid off and building these case studies.

MH

"The clarity of her designs, in spite of the underlying data science and machine learning complexity, is impressive and has greatly contributed to the success of our products. Her design solutions are rooted in a deep understanding of the purpose of the product."

— MARCUS HORBACH, PRINCIPAL DATA SCIENTIST

AG

"During a User Acceptance Test session, Anuja observed me navigating the screen. I was highly impressed with Anuja's approach. Her design was clean, intuitive, and clearly addressed the needs of users across different skill levels."

— ANITA GEORGE, PRINCIPAL ACCOUNT TECH STRATEGIST

WHAT I'D PUSH HARDER FOR

Earlier Working Sessions with the Data Scientist

Live Brainstorming from Week One: If I had started the live working sessions — paper, pen, sketching inside my design software on video calls — from week one instead of halfway through, we would have saved sprint cycles. Brainstorming together inside my tool, not as separate handoffs, was the breakthrough workflow.

FUTURE PLANS

Simplify Even Further

There's always room for improvement: Graying out incompatible models before users ever try them. Simplifying the results screen even more. Automatically letting users enter run model mode from training — no context tab switching. I pushed hard for what mattered. When technical constraints blocked something, that was technical reality, not lack of conviction.



UP NEXT

Next Case Study

003

Up Next →

IQ Plugin

DSML DISCOVERY HUB

AI powered HUB to meet business needs.



(/work/iq-plugin)